

Machine learning methods for continuous glucose monitoring data and their potential in cardiovascular risk assessment

PhD dissertation

Helene Bei Thomsen

Health
Aarhus University
2026



Supervisors and assessment committee

PhD student:

Helene Bei Thomsen

Department of Public Health, Aarhus University, Denmark

Supervisors:

Associate Professor Adam Hulman, PhD (main supervisor)

Department of Public Health, Aarhus University, Denmark

Steno Diabetes Center Aarhus, Denmark

Anders Aasted Isaksen, MD, PhD

Steno Diabetes Center Aarhus, Denmark

Guy Fagherazzi, PhD

Director of the Department of Precision Health

Head of the Deep Digital Phenotyping Research Unit

Luxembourg Institute of Health, Luxembourg

Signe Toft Andersen, MD, PhD

Steno Diabetes Center Aarhus, Denmark

Gødstrup Hospital, Denmark

Assessment committee:

Associate Professor Simon Tilma Vistisen (chairman)

Department of Clinical Medicine, Aarhus University, Denmark **Professor Cristina Soguero-Ruíz**

Department of Signal Theory and Communications, King Juan Carlos University, Spain

Associate Professor Simon Lebech Cichosz

Department of Health Science and Technology, The Faculty of Medicine, Medical Informatics and Image Analysis, Aalborg University, Denmark

Overview

- ☒ A cover
- ☒ TOC
- ☒ List of original papers
- ☐ Introduction
- ☐ Aims and Assumptions
- ☐ Methods - Presentation
- ☐ Methods - Critical evaluation
- ☐ Results
- ☐ Summary - Reflective - Impact (knowledge added)
- ☐ Conclusion
- ☐ Discussion of future aspects
- ☐ References
- ☐ Abstract
 - ☐ Danish
 - ☐ English
- ☐ Papers
- ☐ Co-authorship forms
- ☐ Appendix

Need to have

- ☐ Check big tables from Results (Table 1.3, Table 1.4) is correct with the published article
- ☐ Locate nr from previous tables and change “-” to “nr” in Table 1.4 to match table Table 1.3
- ☐ Ensret OR i tabellerne (Table 1.4)
- ☐ Delete the multimedia part in the footnote of the table after checking what information they link back to and decide if this should be added to supplementary of the dissertation (Table 1.4).
- ☐ Add Adjustments to Table 1.4 or include in supplementary

Formatting

- ☐ Add references to all citations
- ☐ Check big and small letters in headers (uppercase/lowercase beginnings)
- ☐ Ensret if I write introducing text below each chapter or not

Nice to have

Results section

- ☐ List of CGM derived metrics found in the scoping review
- ☐ Add a row “Number of CVD outcomes”- only if time permits and add information to the text as well (Table 1.1).
- ☐ Format this table nicer (Table 1.3), especially the newline parts.

1.1 Study I: Results from a Scoping Review

1.1.1 Study Selection

The study selection process is summarized in Figure 1.1. The Medline and Embase search resulted in 4884 records screened on title and abstract, and 81 reports were assessed for eligibility based on full-text review by two independent reviewers. The citation search on the included 51 reports resulted in 2 additional reports, which finally resulted in 53 included reports in the review. All included and excluded reports are reported in Appendix B.

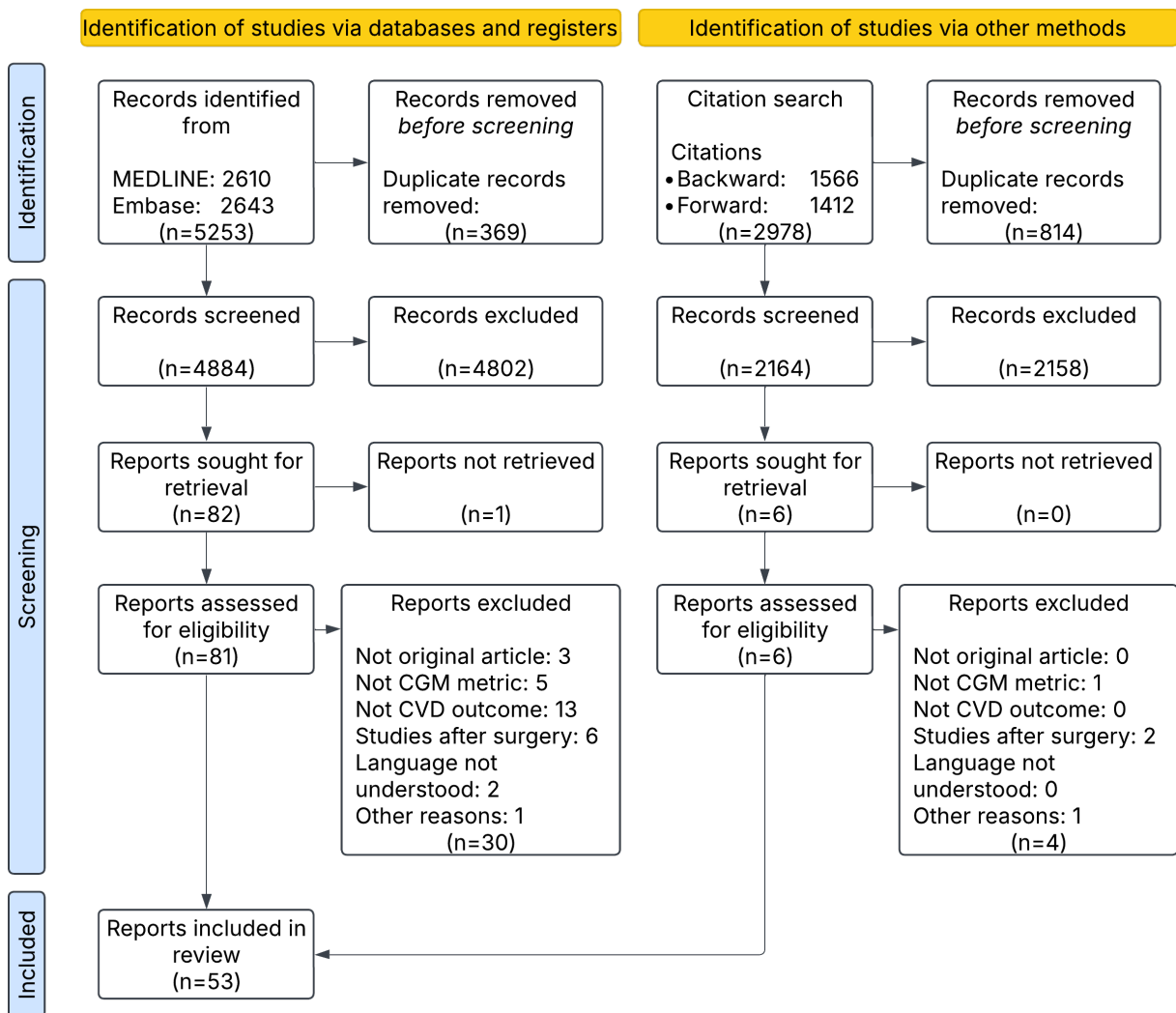


Figure 1.1: Flow diagram of study selection

1.1.2 Study Populations

The study population most common was people with T2D (34 of studies), then people with T1D (20 of 53) and lastly prediabetes (4 of 53). Most studies included adult participants, whereas seven focused on children younger than 18 years. Two studies included only male participants, whereas the remaining studies included both male and female participants.

The geographic distribution was found based on the hospitals and centers where data was collected or by the first author's affiliation when the other option was not available. The distribution was uneven as shown in Figure 1.2. Most studies were conducted in Asia (27 of 53) or Europe (22 of 53). Three studies included data from North America, one was conducted in Australia, and only one included data from an African country, although its study population was predominantly White.^{Borg2011?} No data was included from South America (Appendix C).

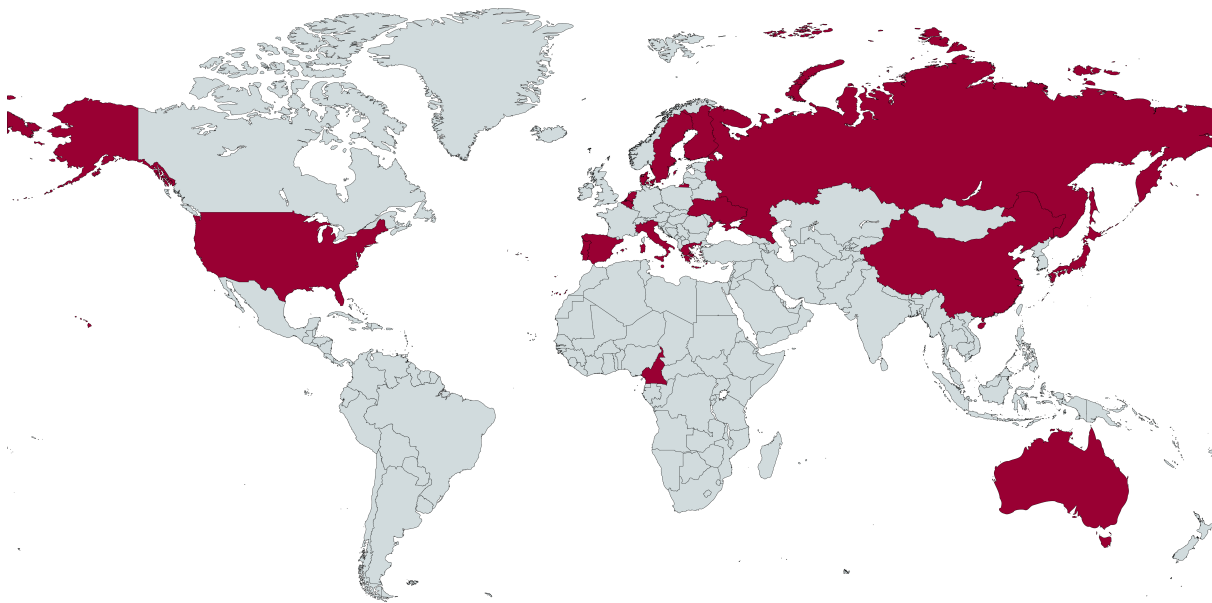


Figure 1.2: Geographical location of the studies. Countries colored in red indicate the geographical locations where the studies were conducted. Map was created using MapChart <https://www.mapchart.net/>.

1.1.3 Study Designs and Demographics

Only one study was a randomized controlled trial,^{33?} and the remaining studies were observational. Most studies used a cross-sectional design, with the cardiovascular outcome assessed at the same time as the CGM sensor was given. Only six studies performed longitudinal analyses. Study sample sizes varied considerably, ranging from 17^{34?} to 6,225^{18?} participants, with a median of 152. CGM data were obtained either from participants' own devices (11 studies) or from CGM devices provided as part of the study (42 studies). Statistical analyses most commonly included linear regression (22 studies) and logistic regression (22 studies), followed by Spearman or Pearson correlation analyses (11 studies) and Cox proportional hazards regression (3 studies). Several studies also performed additional exploratory analyses, such as correlation analyses, to identify relevant covariates for inclusion in regression models.

All included studies primarily investigated associations between CGM-derived measures and cardiovascular outcomes. Only three studies reported predictive performance measures as part of secondary analyses,^{Koroleva?} and none applied machine learning methods. Furthermore, none of the included studies provided openly available data or source code. Instead, availability was either not addressed or limited to access upon reasonable request from the corresponding author.

1.1.4 Continuous Glucose Monitoring

Medtronic CGM devices were the most frequently used (28 of 53 studies), followed by Abbott (9 of 53) and Dexcom (5 of 53). Other manufacturers included Menarini, Meiqi Company, and SIBIONICS. Sampling intervals ranged from 3 to 15 minutes but were reported in only a minority of studies (21 of 53). TIR (22 studies) and MAGE (23 studies) were the most frequently investigated CGM-derived metrics (Table 1.1). When stratified by T1D and T2D then TIR was the most frequently investigated metric in T1D populations (10 of 23 studies), whereas MAGE was less commonly investigated (5 of 22 studies). In T2D populations, TIR and MAGE were investigated in 11 of 23 and 14 of 22 studies, respectively. Studies conducted in T2D populations also tended to include larger sample sizes than those in T1D populations (median [IQR]: TIR, 510 [405–600] vs. 214 [119–547]; MAGE, 251 [89–411] vs. 57 [30–215]).

Table 1.1: Summary of the popularity of the seven most used CGM-derived metrics

CGM-derived Metrics:	TIR	MAGE	MBG	SD	CV	TBR	TAR
Number of Studies, n	23	22	21	19	19	16	15
T1D, n	10	5	9	7	8	9	8
T2D, n	11	14	8	10	8	7	7
Mixed*	2	3	4	2	3	0	0
Total study size for the CGM-derived metrics**	445 [157-577]	156 [63-344]	105 [71-427]	267 [77-480]	342 [98-577]	123 [77-312]	152 [77-425]
T1D**	214 [119-547]	57 [30-215]	105 [31-347]	267 [54-515]	152 [91-515]	124 [102-161]	138 [98-188]
T2D**	510 [405-600]	251 [89-411]	405 [77-499]	371 [95-484]	499 [358-632]	121 [75-499]	405 [75-499]

* Mixed includes both T1D, T2D and prediabetes

** Mixed includes both T1D, T2D and prediabetes

Abbreviations: T1D: type 1 diabetes, T2D: type 2 diabetes, MAGE: mean amplitude of glycemic excursions, TIR: time in range, MBG: mean blood glucose, SD: standard deviation, CV: coefficient of variation, TBR: time below range, TAR: time above range.

1.1.5 Prediction studies

Only three studies evaluated the predictive performance of CGM-derived metrics for cardiovascular outcomes, all using logistic regression models with MAGE as the predictor. Two studies reported receiver operating characteristic area under the curve (ROC-AUC) values of 0.61 and 0.62, respectively,^{43?} and both found MAGE to outperform HbA1c (ROC-AUC: 0.55 and 0.58, respectively). The third study used receiver operating characteristic analysis to determine the optimal threshold for dichotomizing MAGE during variable selection but did not report the ROC-AUC.^{44?} Instead, a threshold of MAGE 3.4 mmol/L was identified as a risk factor for stenosis and/or occlusion, with a sensitivity of 0.60 and a specificity of 0.61.

1.1.6 Association studies

An overview of the different outcomes found in the literature both clinical and subclinical are shown in Table 1.2.

Table 1.2: Summary of established clinical and subclinical outcomes and corresponding abbreviations identified in the literature.

Clinical	Subclinical
CVD = Cardiovascular disease	ABI = Ankle-brachial index
- Macrovascular complications	BP = Blood pressure
- Diabetic macroangiopathy	Carotid artery distensibility = Carotid artery distensibility or carotid distensibility coefficient
CAD = Coronary artery disease	CCA = Common carotid artery
- CHD = Coronary heart disease	CIMT = Carotid artery intima-media thickness
- MI = Myocardial infarction	DBP = Diastolic blood pressure
- Heart attack	FMD = Flow-mediated dilation of the brachial artery
- Heart failure	GSM = Gray-scale median of the carotid arteries
- (Unstable) angina pectoris	HRV = Heart rate variability
Cardiovascular events	IMT = Intima-media thickness
- Ischemic heart disease (IHD)	PWV = Pulse wave velocity
LEAD = Lower extremity artery disease	- ba-PWV = Brachial-ankle pulse wave velocity
MACE = Major adverse cardiovascular events	- cr-PWV = Carotid-radial pulse wave velocity
PAD = Peripheral artery disease	- cf-PWV = Carotid-femoral pulse wave velocity
Stroke = Stroke, Cerebrovascular disease, or Cerebrovascular accident	- Aortic PWV = Aortic pulse wave velocity
	SBP = Systolic blood pressure

1.1.7 Clinical Association Studies

Most studies focused exclusively on subclinical cardiovascular outcomes (37 of 53 studies, 70%), whereas 13 studies (25%) investigated only clinical cardiovascular outcomes and three studies (6%) included both clinical and subclinical outcomes. Because the subsequent chapters of this dissertation focus on clinical cardiovascular outcomes, only these findings are discussed in detail below (Table 1.3 and Table 1.4). A summary of studies investigating subclinical cardiovascular outcomes is provided in Appendix D.

Table 1.3: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Chen, 2020 [†]	Longitudinal (prospective) and cross-sectional (retrospective) FU: in-hospital or within 3 months after discharge from hospital	T2D with CAD (Only male) BG control, n = 90 BG fluctuation, n = 120	210	BG controls: 55.53 ± 7.30 BG fluctuation: 56.41 ± 7.67	BG controls: 6.59 ± 2.30 BG fluctuation group: 6.92 ± 2.25	nr	2 days
He, 2023	Longitudinal (prospective) FU: 1 year	T2D with kidney disease on hemodialysis High TIR, n = 12 Low TIR, n = 15	27	High TIR: 66 [63, 73] Low TIR: 70 [64, 75]	High TIR: 2 [1.7, 10] Low TIR: 5 [0.75, 20]	High TIR: 43 [37, 51] mmol/mol (6.1 [5.5, 6.8]%) Low TIR: 66 [46, 70] mmol/mol (8.2 [6.4, 8.6]%)	14 days
Lu, 2021	Longitudinal (prospective) FU: until death occurred or 3–13 years Median: 6.9 years	T2D, hospitalized	6225	mean age = 61.7 years	mean: 9.7 years	74.0 ± 24.0 mmol/mol (8.9 ± 2.2%)	72 hours
Wei, 2019	Longitudinal (prospective) Median FU [IQR]: 31 months (22, 56)	T2D Divided into three groups: 1) No hypo, n = 1173 2) Mild hypo (lv1), n = 323 3) Severe hypo (lv13), n = 24	1520	No hypoglycemia: 58.59 ± 11.26 Hypoglycemia: 62.27 ± 11.58	No hypoglycemia: 6.46 ± 6.00 Hypoglycemia: 7.78 ± 7.37	No hypoglycemia: (8.19 ± 2.10%) Hypoglycemia: (7.73 ± 1.96%)	3 days

Table 1.3: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Bezerra, 2023	Cross-sectional	T1D	161	37.4 ± 13.4	17.7 ± 10.6	$(7.5 \pm 1.1\%)$	14 days
De Meulemeester, 2024 [†]	Cross-sectional	T1D	808	44.8 ± 15.2	23.1 ± 13.6	63 ± 13 mmol/mol ($7.9 \pm 1.2\%$)	2 weeks
Deng, 2023	Cross-sectional	T2D	860	HP 1 carriers: 53.5 ± 13.3 HP 2-2: 51.7 ± 14.6	HP 1 carriers: 8.6 ± 6.4 HP 2-2: 8.6 ± 6.7	HP1 carriers: 73.0 ± 24.0 mmol/mol ($8.8 \pm 2.2\%$) Hp2-2: 72.0 ± 23.0 mmol/mol (8.7 ± 2.1)	3 days
El Malahi, 2022	Cross-sectional	T1D starting on sensor-augmented pump therapy	515	Mean $\pm$ SD [years]: 42.2 ± 12.5	22.3 ± 11.6	60 ± 9.8 mmol/mol ($7.6 \pm 0.9\%$)	2 weeks
Guo, 2021	Cross-sectional	T1D or T2D with atrial fibrillation With Stroke, n = 48 Without Stroke, n = 462	510	Stroke: 70.3 ± 12.1 No stroke: 68.1 ± 9.4	nr	No Stroke: 7.4 ± 2.1 Stroke: 8.2 ± 1.7	72 hour
Li, 2020	Cross-sectional	T2D with LEAD, n = 179 T2D without LEAD, n = 157	336	Without LEAD: 55.94 ± 12.45 With LEAD: 65.56 ± 11.99	Without LEAD: 6.92 ± 3.54 With LEAD: 10.32 ± 4.14	Without LEAD: $7.85 \pm 1.41\%$ With LEAD: $8.97 \pm 1.63\%$	72 hours

Table 1.3: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Magri, 2018 [†]	Cross-sectional	T2D	121	64 [57, 68]	3 [2, 5]	median: 45 mmol/mol (6.8%)	72 hours
Mi, 2012	Cross-sectional	T2D with chest pain Without CAD, n = 202 With CAD, n = 84	286	Without CAD: 62.8 ± 8.7 With CAD: 66.6 ± 9.2	nr	Non-CAD: 7.51 ± 0.80% CAD: 7.75 ± 0.92%	72 hours (Only intermediate 48 hours were used)
Sheng, 2023	Cross-sectional	T2D, hospitalized	545	mean age of (61.22 ± 11.21) years	nr	8.51 ± 1.85%	7-14 days
Su, 2011	Cross-sectional	T2D with chest pain Without CAD, n = 92 With CAD, n = 252	344	Without CAD: 61 ± 9 With CAD: 65 ± 9	Without CAD: 4.8 ± 5.7 With CAD: 6.5 ± 6.4	non-CAD: 7.5 ± 1.4% CAD: 7.6 ± 1.5%	72 hours (only 48 hours used)
Watanabe, 2017	Cross-sectional	Prediabetes, hospitalized	28	64.3 ± 12.8	nr	5.41 ± 0.35%	72 hours (Only used the middle 48 hours)

Zhang, 2013	Cross-sectional	T2D with cardiovascular complications Group A: healthy individuals Group B: T2D without cardiovascular complications Group C: T2D with cardiovascular complications	92	Group A: 56.3 ± 6.1 Group B: 56.1 ± 6.6 Group C: 61.7 ± 7.2	nr	Group A: 5.3 ± 0.3% Group B: 6.6 ± 1.2% Group C: 7.5 ± 1.4%	72 hours
-------------	-----------------	---	----	---	----	---	----------

[†] This study appears both in the clinical and subclinical disease outcome tables due to investigating multiple CVD outcomes. Note: Age is reported in years either as an interval, mean ± SD, or median (IQR). Diabetes duration values originally reported in months were converted to years for consistency (months ÷ 12).
Abbreviations: CGM: continuous glucose monitoring, FU: follow-up, T1D: type 1 diabetes, T2D: type 2 diabetes, CAD: coronary artery disease, hypo: hypoglycemia, LEAD: lower extremity arterial disease, IQR: interquartile range, nr: not reported.

Clinical cardiovascular outcomes comprised cardiovascular mortality, major adverse cardiovascular events, macrovascular complications, coronary artery disease, stroke, and peripheral artery disease. Most studies were observational and differed substantially in study populations, CGM-derived metrics, outcome definitions, and adjustment strategies. Consequently, direct comparison across studies was challenging. The reported effect estimates from the least and most adjusted analyses are summarized in Table 1.4.

Table 1.4: : Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
Cardiovascular Mortality					
Lu et al ² , 2021					Model 1: Demographic
TIR > 85%	HR 1.00	< 0.001 (trend)	HR 1.00	0.02 (trend)	Model 2: Demographic
TIR 71% – 85%	HR: 1.43 (0.95–2.14)	< 0.001 (trend)	HR: 1.35 (0.90–2.04)	0.02 (trend)	Anthropometric
TIR 51% – 70%	HR: 1.66 (1.12–2.45)	< 0.001 (trend)	HR: 1.47 (0.99–2.19)	0.02 (trend)	Lifestyle
TIR ≥ 50%	HR: 2.15 (1.47–3.13)	< 0.001 (trend)	HR: 1.85 (1.25–2.72)	0.02 (trend)	Medical History
TIR as a continuous variable (each 10% decrease)	HR 1.08 (1.03–1.13)	—	HR 1.05 (1.00–1.11)	—	Physiological Biochemical Medication
Wei et al ^{Wei2019?*} , 2019					Least adjusted: Unadjusted
Hypoglycemic events	HR: 2.03 (1.21–3.41)	—	HR: 2.64 (1.40–4.99)	—	Most adjusted: Demographic Anthropometric Lifestyle Medical History Biochemical Medication
Major Adverse Cardiovascular Events					
He et al ^{He2023?} , 2023					
Blood glucose risk index	HR: 0.97 (0.85–1.10)	0.61	HR: 0.98 (0.85–1.13)	0.75	
Low blood glucose index	HR: 2.37 (1.16–4.83)	0.02	HR: 2.73(1.21–6.16)	0.02	

Table 1.4: : Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
High blood glucose index	HR: 0.94 (0.81–1.08)	0.38	HR: 0.94 (0.81–1.09)	0.44	
Average of daily risk range	HR: 1.00 (0.93–1.07)	> 0.99	HR: 1.01 (0.93–1.09)	0.80	
GMI	HR: 0.98 (0.91–1.06)	0.65	HR: 0.99(0.91–1.07)	0.78	
M-value	HR: 0.98 (0.91–1.05)	0.54	HR: 0.98(0.91–1.06)	0.64	
Wei et al ^{Wei2019?*} , 2019					
Hypoglycemic events	HR: 1.501 (1.207–1.866)	—	HR: 1.615 (1.239–2.106)	< 0.001	
Macrovascular Complications					
De Meulemeester et al ^{Meulemeester2024?†,*} , 2024					
TIR	OR: 0.939 (0.829–1.063)	> 0.05	OR: 0.896 (0.738–1.087)	> 0.05	
TITR	OR: 0.901 (0.775–1.047)	> 0.05	OR: 0.933 (0.745–1.169)	> 0.05	
Bezerra et al ^{Bezerra2023?} , 2023					
TIR	OR: 0.66 (0.46–0.93)	0.02	OR: 0.68 (0.39–1.16)	0.15	
Time below 54 mg/dL	OR: 1.10 (0.88–1.38)	0.39	OR: 0.92 (0.62–1.34)	0.65	
TBR	OR: 0.93 (0.80–1.09)	0.39	OR: 0.77 (0.54–1.11)	0.17	
TAR	OR: 1.04 (1.01–1.08)	0.01	OR: 1.04 (0.99–1.10)	0.08	
Time above 250 mg/dL	OR: 1.04 (1.00–1.08)	0.03	OR: 1.03 (0.97–1.09)	0.29	
CV	OR: 1.03 (0.95–1.11)	0.52	OR: 0.92 (0.81–1.06)	0.25	
GMI	OR: 2.17 (1.14–4.11)	0.02	OR: 2.03 (0.77–5.37)	0.15	
Deng et al ^{Deng2023?} , 2023					
% CV tertile 1 (Hp1; reference)	OR: 1.000	0.07 (interaction)	OR: 1.000	0.008 (interaction)	

Table 1.4: : Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
% CV tertile 1 (Hp2-2; reference)	OR: 1.000	0.07 (interaction)	OR: 1.000	0.008 (interaction)	
% CV tertile 2 (Hp1)	OR: 1.483 (0.907–2.423)	0.12	OR: 1.048 (0.528–2.078)	0.89	
% CV tertile 2 (Hp2-2)	OR: 1.399 (0.829–2.358)	0.21	OR: 0.659 (0.296–1.466)	0.31	
% CV tertile 3 (Hp1)	OR: 2.347 (1.393–3.957)	0.001	OR: 2.461 (1.183–5.121)	0.02	
% CV tertile 3 (Hp2-2)	OR: 1.217 (0.731–2.027)	0.45	OR: 0.540 (0.245–1.191)	0.13	
El Malahi et al ^{Malahi2022?} , 2022					
TIR	—	—	—	> 0.05	
SD	—	—	—	> 0.05	
CV	—	—	—	> 0.05	
Magri et al ^{3†} , 2018					
TBR	—	—	OR: 1.12 (1.014–1.228)	0.02	
Lowest BG value	—	—	—	—	
Area under the TBR curve	—	—	—	—	
Coronary Artery Disease					
De Meulemeester et al ^{Meulemeester2024?†,*} , 2024					
TITR	OR: 1.039 (0.812–1.330)	> 0.05	OR: 1.255 (0.874–1.803)	> 0.05	
TIR	OR: 1.072 (0.866–1.328)	> 0.05	OR: 1.164 (0.844–1.607)	> 0.05	

Table 1.4: : Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
Sheng et al ^{Sheng2023?} , 2023					
TIR < 20%	—	—	OR: 2.143 (1.554–3.287)	—	
TIR 20% – 40%	—	—	OR: 1.049 (0.945–2.022)	—	
TIR 40% – 60%	—	—	OR 0.854 (0.495–1.473)	—	
TIR 60% – 80%	—	—	OR: 0.617 (0.423–1.312)	—	
TIR > 80%	—	—	OR: 0.470 (0.143–1.545)	—	
Chen et al ^{Chen2020?†,*} , 2020					
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L versus high BG fluctuations (angina pectoris)	—	—	$\chi^2 = 5.797$	0.02	
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L versus high BG fluctuations (angina pectoris)	—	—	$\chi^2 = 7.490$	0.006	
Wei et al ^{Wei2019?*} , 2019					

Table 1.4: : Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
Hypoglycemic events (myocardial Infarction)	HR: 1.901 (1.067–3.389)	—	HR: 1.549 (0.768–3.124)	0.03	
Hypoglycemic events (unstable angina pectoris)	HR: 1.226 (0.857–1.753)	—	HR: 1.218 (0.794–1.869)	0.30	
Shu-Hua et al ^{Hua?e} , 2012					
MAGE level (≥ 3.4 mmol/L)	—	—	OR: 2.286 (1.176–4.446)	0.02	
Su et al ^{1*} , 2011					
MAGE ≥ 3.4 mmol/L	—	—	OR: 2.612 (1.423–4.831)	0.002	
MAGE	—	—	AUC: 0.618 (0.555–0.680)	0.001	
Gensini Score					
Chen et al ^{Chen2020?†,*} , 2020					
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L	—	—	$t = 6.210$	< 0.001	
Watanabe et al ^{Watanabe2017?*} , 2017					
MAGE	—	—	$r = 0.742$	< 0.001	
Shu-Hua et al ^{Hua2012?*} , 2012					
			Unstandardized coefficient $\beta = 4.817$; SE = 1.164;		
MAGE	—	—		0.003	

Table 1.4: : Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
			Standardized coefficient $t = 2.984$		
Su et al ^{1*} , 2011					
MAGE	—	—	Unstandardized $\beta = 7.010$; SE = 1.466; standardized $\beta = 0.237$; $t = 4.783$	< 0.001	
Syntax Score					
Watanabe et al ^{Watanabe2017?*} , 2017					
MAGE	—	—	$r = 0.776$	< 0.001	
Zhang et al ^{4*} , 2013					
MAGE	—	—	$r = 0.518$	0.01	
BG fluctuations from 00:00 to 03:00	—	—	$r = -0.442$	0.04	
BG fluctuations from 03:00 to 06:00	—	—	$r = -0.208$	0.34	
BG fluctuations from 06:00 to 08:00	—	—	$r = 0.678$	< 0.001	
BG fluctuations from 08:00 to 11:00	—	—	$r = 0.115$	0.60	
BG fluctuations from 11:00 to 13:00	—	—	$r = 0.523$	0.01	
BG fluctuations from 13:00 to 17:00	—	—	$r = 0.257$	0.24	
BG fluctuations from 17:00 to 19:00	—	—	$r = 0.358$	0.09	
BG fluctuations from 19:00 to 24:00	—	—	$r = -0.018$	0.93	
Stroke					

Table 1.4: : Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
De Meulemeester et al Meulemeester2024? ^{†,*} , 2024					
TITR	OR 0.651 (0.470–0.902)	< 0.05	OR 0.546 (0.347–0.858)	< 0.01	
TIR	OR 0.749 (0.588–0.955)	< 0.05	OR 0.617 (0.440–0.866)	< 0.01	
Guo et al Guo2021? [*] , 2021					
TIR: Q1 (≤ 46%; reference)	OR: 1.00	< 0.001	OR: 1.00	< 0.001	
TIR: Q2 (46% – 65%)	OR: 0.86 (0.72–0.95)	< 0.001	OR: 0.80 (0.68–0.92)	< 0.001	
TIR: Q3 (65% – 81%)	OR: 0.71 (0.61–0.81)	< 0.001	OR: 0.64 (0.53–0.79)	< 0.001	
TIR: Q4 (> 81%)	OR: 0.66 (0.58–0.80)	< 0.001	OR: 0.59 (0.50–0.74)	< 0.001	
TIR (per 10% increase)	OR: 0.93 (0.85–0.98)	0.008	OR: 0.89 (0.82–0.95)	< 0.001	
Wei et al Wei2019? [*] , 2019					
Hypoglycemic events	HR: 1.691 (1.144–2.499)	—	HR: 1.813 (1.110–2.960)	0.06	
Peripheral artery disease					
De Meulemeester et al Meulemeester2024? ^{†,*} , 2024					
TITR	OR 0.680 (0.426–1.085)	> 0.05	OR 0.807 (0.382–1.703)	> 0.05	
TIR	OR 0.736 (0.520–1.042)	> 0.05	OR 0.811 (0.470–1.398)	> 0.05	
Lower Extremity Arterial Disease					
Li et al Li2020? [*] , 2020					

Table 1.4: : Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
TIR	OR: 0.979 (0.968–0.991)	< 0.001	OR: 0.979 (0.965–0.992)	0.002	
CV	OR: 1.040 (1.003–1.078)	0.04	OR: 1.038 (0.996–1.081)	0.08	
SD	OR: 1.325 (1.038–1.691)	0.02	OR: 1.158 (0.824–1.627)	0.40	
TIR-without LEAD (1)	OR: 1.00	—	OR: 1.00	—	
TIR-mild LEAD (1)	OR: 0.98 (0.97–1.00)	0.14	OR: 0.99 (0.97–1.01)	0.25	
TIR-moderate LEAD (1)	OR: 0.97 (0.95–0.99)	0.007	OR: 0.97 (0.95–0.99)	0.01	
TIR-without severe LEAD (1)	OR: 0.96 (0.94–0.98)	0.002	OR: 0.96 (0.94–0.98)	0.003	
CV-without LEAD	—	—	OR: 1.00	—	
CV-mild LEAD	—	—	OR: 1.03 (0.98–1.07)	0.28	
CV-moderate LEAD	—	—	OR: 1.02 (0.96–1.09)	0.48	
CV-without severe LEAD	—	—	OR: 1.02 (0.95–1.09)	0.60	
TIR-without LEAD (2)	—	—	OR: 1.00	—	
TIR-mild LEAD (2)	—	—	OR: 0.97 (0.96–1.08)	0.06	
TIR-moderate LEAD (2)	—	—	OR: 0.98 (0.95–0.99)	0.01	
TIR-without severe LEAD (2)	—	—	OR: 0.97 (0.95–0.99)	0.02	
SD-without LEAD	—	—	OR: 1.00	—	
SD-mild LEAD	—	—	OR: 0.88 (0.47–1.64)	0.69	
SD-moderate LEAD	—	—	OR: 1.28 (0.58–3.07)	0.58	
SD-without severe LEAD	—	—	OR: 1.52 (0.92–2.41)	0.10	

All hazard ratios (HRs), odds ratios (ORs), and areas under the curve (AUCs) are reported as follows: point estimate (95% CI).

* This study appears multiple times as it investigated multiple cardiovascular disease outcomes.

† This study appears in both the clinical and subclinical disease outcome tables owing to the investigation of multiple cardiovascular disease outcomes.

The full table with adjustments is provided in **Multimedia Appendix 10**. Further elaboration on the adjusted variables can be found in **Multimedia Appendices 7 and 8**.

Abbreviations: nr: Not applicable or not available/not reported, TIR: time in range, GMI: glucose management indicator, TITR: time in tight range, TBR: time below range, TAR: time above range, CV: coefficient of variation, BG: blood glucose, MAGE: mean amplitude of glycemic excursions, LAGE: largest amplitude of glycemic excursions, MODD: mean of daily differences, LEAD: lower extremity arterial disease.

References

1. Su G, Mi S, Tao H, et al. Association of glycemic variability and the presence and severity of coronary artery disease in patients with type 2 diabetes. *Cardiovascular diabetology* 2011;10(1):19–9.
2. Lu J, Wang C, Shen Y, et al. Time in range in relation to all-cause and cardiovascular mortality in patients with type 2 diabetes: A prospective cohort study. *Diabetes care* 2021;44(2):549–55.
3. Magri CJ, Mintoff D, Camilleri L, Xuereb RG, Galea J, Fava S. Relationship of hyperglycaemia, hypoglycaemia, and glucose variability to atherosclerotic disease in type 2 diabetes. *Journal of Diabetes Research* [Internet] 2018;2018(1):7464320. Available from: <https://onlinelibrary.wiley.com/doi/abs/10.1155/2018/7464320>
4. Zhang X, Xu X, Jiao X, et al. The effects of glucose fluctuation on the severity of coronary artery disease in type 2 diabetes mellitus. *Journal of diabetes research* 2013;2013(2013):1–6.

Study I: Search strings A

Table A.1: Search in Ovid MEDLINE (Search date: 22 January 2024)

#	Searches	Results
1	Blood Glucose	10241
2	Self-Monitoring/ Continuous Glucose Monitoring/	34
3	((blood-glucose or bloodglucose or glucose) adj2 (monitor\$ or selfmonitor)) <i>or</i> cgmorb _{gmor} iscgmorr _{tcgmor} (<i>timeadj3range</i>)). <i>ti, ab, kf, kw</i> or bloodsugar\$ or glucose\$ or glyc?emic\$ or sugar). <i>ti, ab, kf, kw.</i>)adj5(<i>flashorfluctuat</i> or intermittent\$ or scan\$ or variabilit). <i>ti, ab, kf, kw.</i> 8669 5 <i>or</i> /1— 4 38366 6 <i>expCardiovascularDiseases/</i> 2760764 7 (cardiovasc or cvd or cardiometabol\$ or cardio-metabol\$ or heart\$ or macrovasc\$ or vasc). <i>ti, ab, kw, kf.</i> 2182473 8 <i>or</i> /6— 7 3971342 9 <i>expDiabetesMellitus/</i> 518864 10 (diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d). <i>ti,ab,kw,kf.</i> or/9-10	815092
11	or/9-10	875527
12	and/5,8,11	2416
13	exp Animals/ not Humans/	5189493
14	12 not 13 Updated search: 11 March 2025	2300
15	limit 14 to (da="20231201-20261231" or dt="20231201-20261231" or ez="20231201-20261231" or ed="20231201-20261231")	310
16	or/14-15	2610

Table A.2: Search in Embase (Ovid) (Search date: 22 January 2024)

#	Searches	Results
1	blood glucose monitoring/	37574
2	exp continuous glucose monitoring system/	5648
3	((blood-glucose or bloodglucose or glucose) adj2 (monitor\$ or selfmonitor))orcmorbmgmorisecgmorrtecmor(timeadj3range)).ti, ab, kf, kw. 4850 or bloodsugar\$ or glucose\$ or glyc?emic\$ or sugar).ti, ab, kf, kw.)adj5(flashorfluctuat or intermittent\$ or scan\$ or variabilit).ti, ab, kf, kw. 16430 5 or/1–4 81510 6 expcardiovascularisease/ 5122163 7 (cardiovasc or cvd or cardiometabol\$ or cardio-metabol\$ or heart\$ or macrovasc\$ or vasc).ti, ab, kw, kf. 3044925 8 or/6–7 6303633 9 expdiabetesmellitus/ 1239184 10 (diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d).ti,ab,kw,kf.	1237019
11	or/9-10	1501812
12	and/5,8,11	11217
13	(rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset\$1).ti. and animal experiment/	1237966
14	animal experiment/ not (human experiment/ or human/)	2601109
15	or/13-14	2672240
16	12 not 15	10861
17	limit 16 to exclude medline journals Updated search: 11 March 2025	2176
18	limit 17 to (dc="20231201-20261231" or dd="20231201-20261231" or rd="20231201-20261231")	467
19	or/17-18	2643

Study I: Included and excluded reports **B**

C Study I: Geographical location

Asia (n=27) • China (n=19) • Japan (n=8) Europe (n=22) • Italy (n=7) • 3 Belgium (n=3) • 2 Ukraine (n=2) • 2 Finland (n=2) • 2 Spain (n=2) • 1 Greece (n=1) • 1 Malta (n=1) • 1 Sweden (n=1) • 1 The Netherlands (n=1) • 1 Portugal (n=1) • 1 Russia (n=1) North America (n=2) • 2 USA (n=2) Australia / Oceania (n=1) • 1 Australia (n=1)

Borg 2011(ADAG study) had study centers in: • Europe: Netherlands, Denmark, Italy • Africa: 2011 • North America: USA

Borg R, Kuenen J C, Carstensen B, Zheng H, Nathan D M, Heine R J, et al. HbA1(c) and mean blood glucose show stronger associations with cardiovascular disease risk factors than do postprandial glycaemia or glucose variability in persons with diabetes: the A1C-Derived Average Glucose (ADAG) study. 2011; ADAG Study Group, editor. Diabetologia. 54: 69–72. doi:10.1007/s00125-010-1918-2

Study I: Subclinical results

D

- ☐ Add table with subclinical study design overviews
- ☐ Add table overview of subclinical results

E Study II: Prediction stability

Prediction stability plots (?@fig-pred__stability) revealed more stability the fewer predictors.

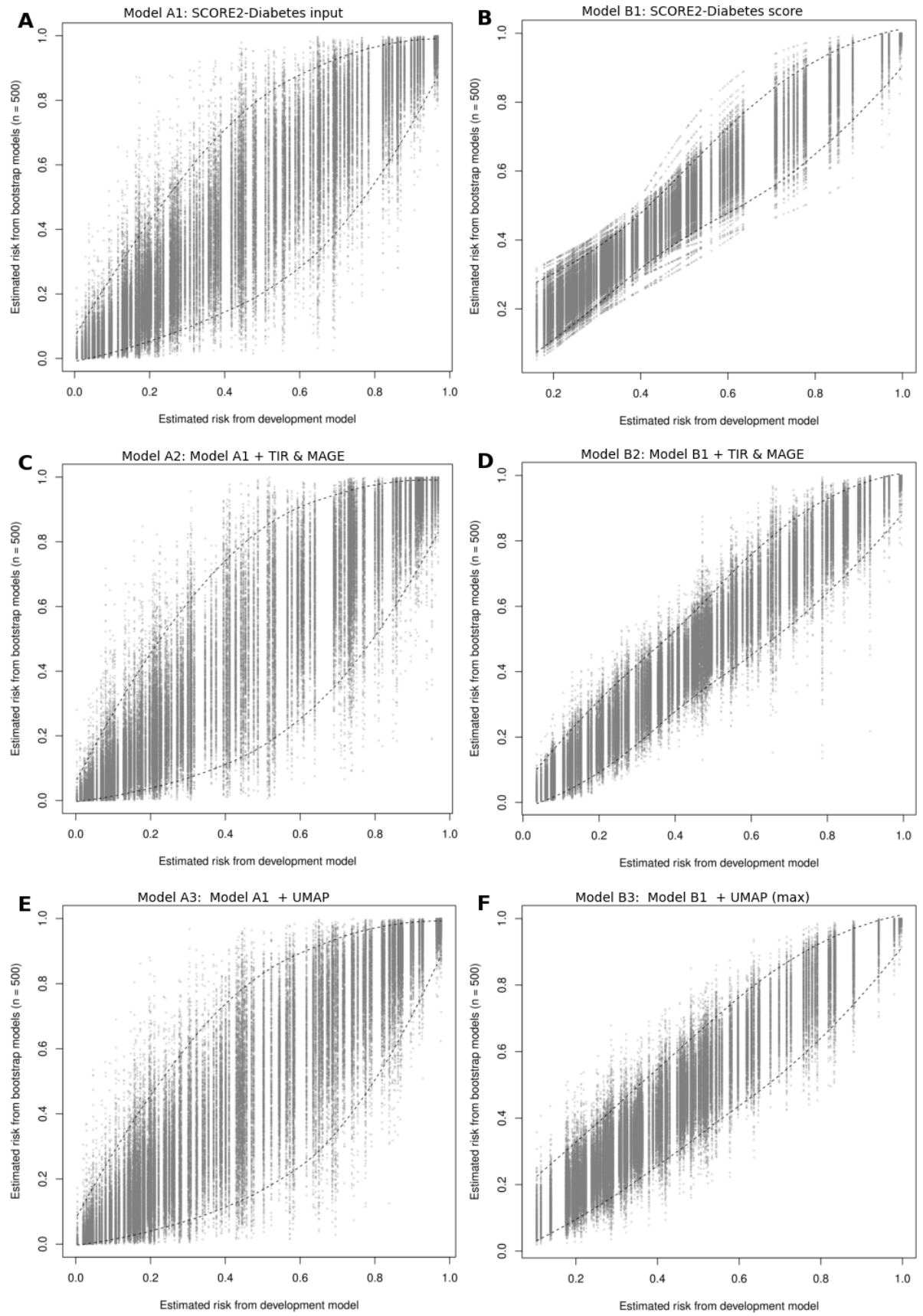


Figure E.1: Prediction stability plots show the predictions (estimated risk) for the full dataset from the development model (equivalent to the apparent model) on the x-axis, and the predictions from all bootstrap models on the y-axis. Tight alignment along the diagonal indicates higher prediction stability.

F Study III: Surveillance Error Grid Results

Table F.1: Results from the Surveillance Error Grid for White participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
White	LOCF	None	69%	69%	69%	69%	69%	69%
		Slight	26%	26%	26%	26%	26%	26%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	28%	28%	28%	28%	28%	28%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Generalized	None	72%	73%	73%	73%	73%	73%
		Slight	25%	25%	25%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Transfer Learned	None	72%	72%	73%	73%	73%	73%
		Slight	25%	24%	24%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

□ maybe add this one to appendix, since I think they show the same things

Table F.2: Results from the Surveillance Error Grid for Black participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
Black	LOCF	None	71%	71%	71%	71%	71%	71%
		Slight	24%	24%	24%	24%	24%	24%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	27%	27%	27%	27%	27%	27%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

Group	Model	Zone	0	20	40	60	80	100
	Base -Generalized	Extreme	0%	0%	0%	0%	0%	0%
		None	75%	74%	75%	75%	74%	75%
		Slight	23%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
	Transfer Learned	Extreme	0%	0%	0%	0%	0%	0%
		None	75%	74%	75%	75%	74%	75%
		Slight	22%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

G Full papers

G.1 Paper I

G.2 Paper II

G.3 Paper III